SHIELD-AI Glossary

Plain-language definitions of the terms used in our briefings

Security Architecture Practice

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Abstract
A short glossary for executives, boards, and program managers meeting AI security vocabulary for the first time.

SHIELD-AI Glossary

Terms are grouped the way you will meet them in a briefing. Definitions are written for a non-technical reader; the technical meaning is included where it matters.

The core concepts

Agentic AI (agent). An AI system that does not just answer questions but takes actions: sending email, editing code, calling APIs, spending money, changing configuration. An agent has tools, and tools can do damage.

Autonomy. How much an agent can do without a human in the loop. Low autonomy: suggests, waits for approval. High autonomy: acts, reports afterwards. Most security incidents involve agents whose autonomy was set higher than their guardrails.

Tool. A capability an agent can call: an API, a database query, a shell command, a web action. Each tool is a potential attack surface. The framework treats tools as the central risk: an agent is only as safe as its tools.

Prompt injection. Hiding instructions inside content an agent reads. A document, a web page, or a comment can contain text that tells the agent to do something the user never asked for. This is the AI equivalent of a phishing email, and it is the most common attack class in the incident ledger.

Sandbox. A contained environment where an agent runs, with limited access to the host system and the network. A sandbox is only as good as its configuration; several major incidents happened because the sandbox was misconfigured, not because it was missing.

Guardrail. A rule or control that constrains what an agent can do. Guardrails sit between the model and the action.

Maturity level (L1-L4). A four-step scale used throughout the framework, consistent with ISM and the Essential Eight: L1 Initial, L2 Managed, L3 Defined, L4 Optimised. It gives a repeatable way to describe posture.

The framework mechanics

Pillar. One of the seven security domains of the framework: Model Security, Identity and Access, Data Security, Tool and Pipeline Security, Monitoring and Incident Response, Governance and Risk, Execution Environment.

Control. A specific, testable security requirement, numbered within its pillar (for example MS-04, TP-10). SHIELD-AI has 59 controls. Each maps to one or more regulatory obligations and to documented incidents.

Policy as code. Expressing security rules as machine-readable code instead of prose. SHIELD-AI controls are written as Rego rules evaluated by an OPA policy engine. The advantage: rules can be tested, versioned, and enforced automatically, and an audit becomes a test run.

Deny-override. The evaluation rule used by the policy engine: if any control says deny, the action is blocked, even if another control allows it. It is the opposite of allow-override, where an exception can quietly weaken security. Deny-override fails closed.

Fail closed. If the security system cannot confirm an action is safe (engine down, config missing, dependency unreachable), the action is blocked. Fail-open, by contrast, lets the action through when the control is unavailable, which is how several real-world breaches happened.

Sidecar. A small security service that runs next to the application and intercepts its traffic. SHIELD-AI's reference implementation deploys as a sidecar, which means existing platforms can be protected without rebuilding them.

Reference implementation. The working example architecture that ships with the framework: the components, diagrams, and policy engine that show how the controls are deployed in practice.

The technology vocabulary

MCP (Model Context Protocol). A standard way for agents to connect to external tools and data. Because MCP servers can execute commands, they have become a favourite attack surface; several 2026 incidents involved exposed or poisoned MCP servers.

Rego. The policy language used by OPA, the open-source policy engine. It is how SHIELD-AI controls are expressed as executable rules.

OPA (Open Policy Agent). An open-source engine that evaluates policy rules. It is the enforcement core of the reference implementation.

LLM (Large Language Model). The model that powers the agent: GPT, Claude, Gemini, and open-weight alternatives. The framework is model-agnostic; the controls protect the system around the model, not the model itself.

RAG (Retrieval-Augmented Generation). Giving a model access to a document store so it can answer from your data. The retrieval step is an injection point: documents can contain hidden instructions.

SBOM (Software Bill of Materials). A list of the components in a piece of software. The framework requires SBOMs for models and dependencies so supply-chain attacks can be traced.

Egress. Outgoing network traffic. Egress controls restrict which external destinations an agent can reach; several evaluation-harness escapes happened because egress was left open.

The regulatory vocabulary

ISM (Information Security Manual). The Australian government's security standard, published by the Australian Cyber Security Centre.

Essential Eight (E8). ACSC's baseline of eight mitigation strategies for cyber security.

SOCI Act. The Security of Critical Infrastructure Act, which imposes obligations on critical infrastructure operators.

CPS 234. APRA's prudential standard on information security for banks, insurers, and superannuation funds.

CPG 230. APRA's guidance on managing information security risk, including third-party and outsourcing risk.

PSPF. The Protective Security Policy Framework, applying to Australian government entities.

Privacy Act / APP 11. The Australian Privacy Act and its Australian Privacy Principle 11, which requires organisations to protect personal information from misuse, loss, and unauthorised access.

ASIC. The Australian Securities and Investments Commission, which has stated expectations for AI governance in financial services.

AU Guardrails. The Australian government's proposed mandatory guardrails for AI in high-risk settings.

Plain-language definitions are for briefings. Where a definition differs from the technical meaning, the framework document is authoritative.